

KnowSelf: Teaching LLM Agents When to Think, Reflect, c

Agentic knowledgeable self-awareness for efficient planning in embodied environments

Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge Injection Hurts Eff

Problem framing:

- Existing paradigms inject trajectories/feedback/knowledge across all states.
- No situational judgment on when external knowledge is necessary.
- Overhead increases while policy quality can degrade.

Three common paradigms:

- 1) Trajectory imitation: follows demonstrations rigidly.
- 2) Trial-and-error reflection: retries extensively.
- 3) Knowledge-augmented planning: always consults KB/long context.

Contrast: Human decision-makers adapt cognitive effort by context.

Trajectory
Imitation

Rigid transfer

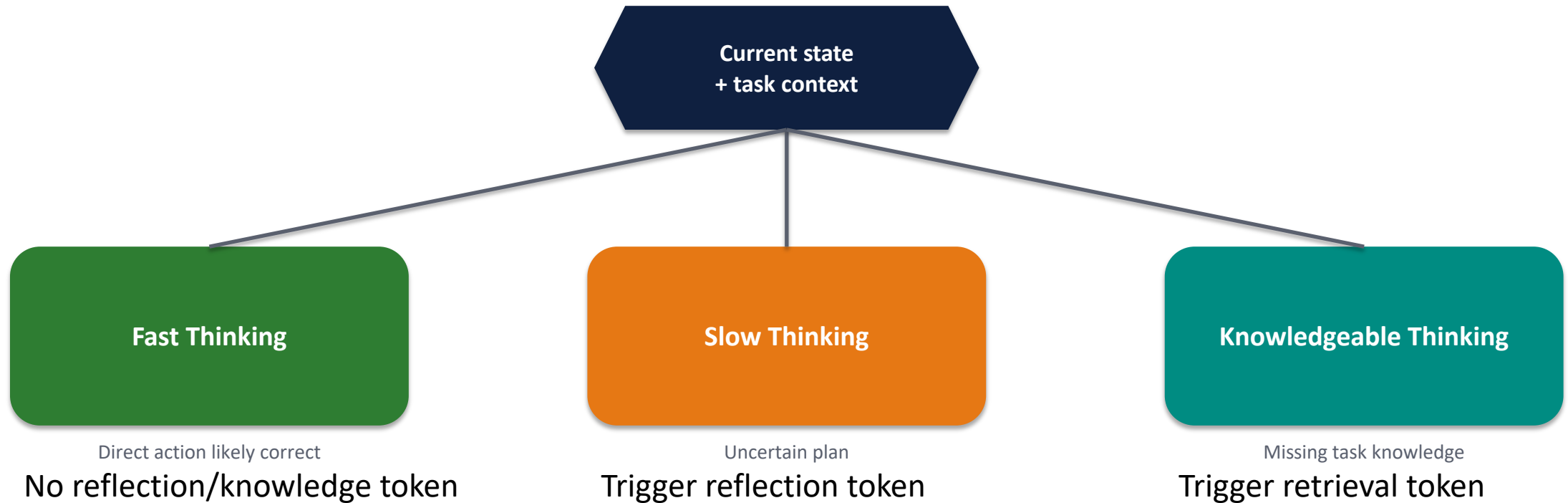
Trial-and-Error
Reflection

High retry cost

Knowledge-
Augmented

Over-querying knowledge

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking



KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

Reference: framework figure (figure_3.jpg)



KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Reference: qualitative examples (figure_1.jpg)

Task A: Put a hot mug in cabinet

✓ KnowSelf: triggers reflection, heats mug then stores it

✗ OpenAI-o1: fails to enforce heated-state prerequisite

Task B: Put a saltshaker in drawer

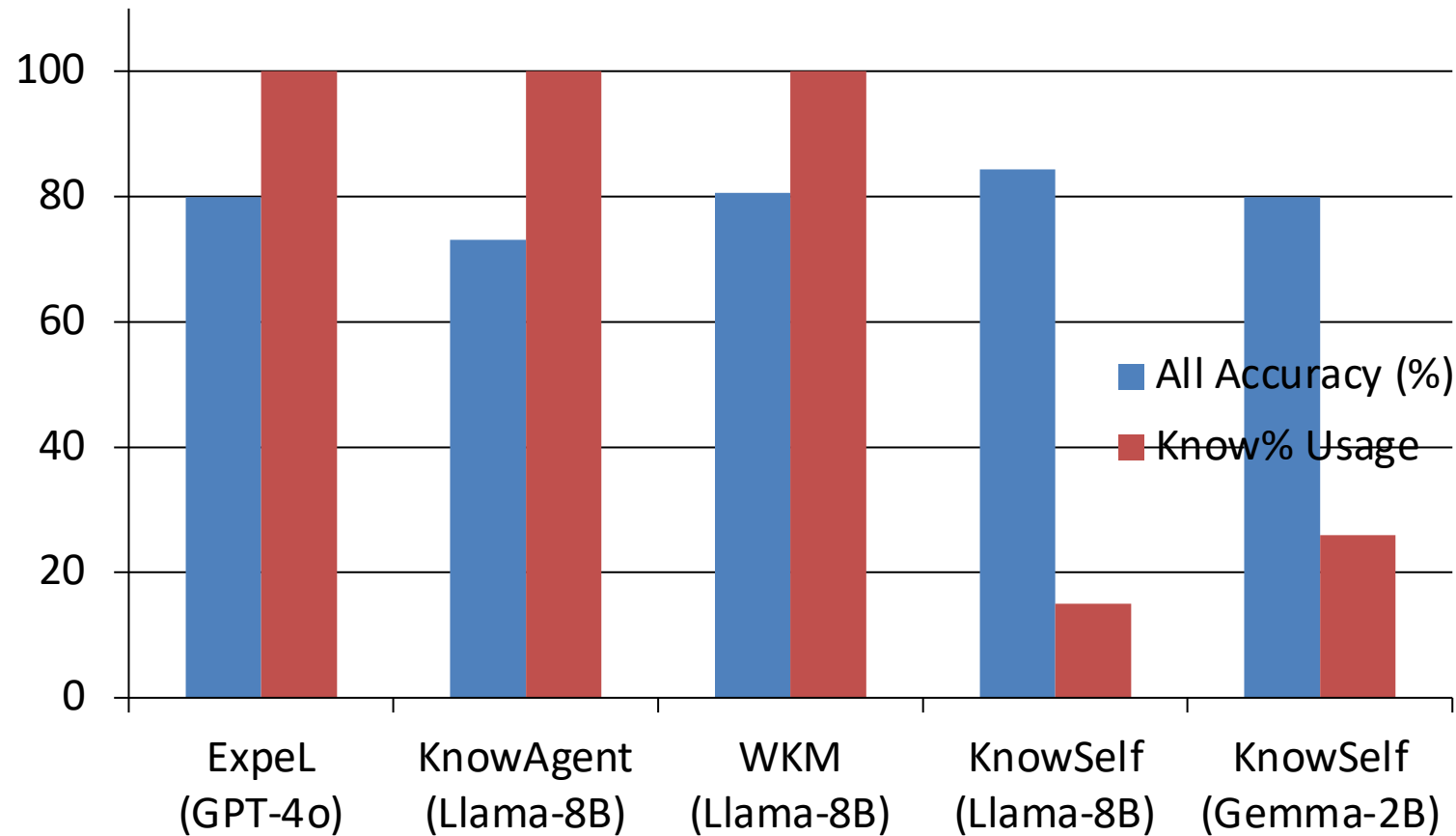
✓ KnowSelf: retrieves object-location knowledge and completes placement

✗ DeepSeek-R1: loops search actions and gets stuck

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Backbone	Method	Know%	pick_cool	pick_heat	look_at	pick_place	pick_clean	All
GPT-4o	ReAct	-	73.81	72.73	81.82	75.00	35.29	76.12
GPT-4o	Reflexion	-	66.67	72.73	86.36	66.67	41.18	72.39
GPT-4o	ExpeL	100	76.19	77.27	81.82	83.33	52.94	79.85
Llama-8B	ReAct	-	71.43	75.00	72.73	66.67	52.94	71.64
Llama-8B	KnowAgent	100	73.81	72.73	72.73	75.00	47.06	73.13
Llama-8B	WKM	100	78.57	79.17	77.27	91.67	58.82	80.60
Llama-8B	KnowSelf	15	80.95	87.50	81.82	91.67	70.59	84.33
Gemma-2B	ReAct	-	69.05	72.73	68.18	66.67	35.29	68.66
Gemma-2B	ETO	-	76.19	68.18	72.73	58.33	52.94	70.15
Gemma-2B	KnowAgent	100	73.81	68.18	72.73	66.67	35.29	70.15
Gemma-2B	WKM	100	78.57	79.17	72.73	83.33	47.06	76.12
Gemma-2B	KnowSelf	26	80.95	77.27	81.82	83.33	52.94	79.85

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage Over Full-Knowledge



Efficiency signal:

- Full-knowledge methods query external knowledge at 100%.
- KnowSelf achieves best score (84.33) at only 15% knowledge on Llama-8B.
- Gemma-2B KnowSelf reaches 79.85 with 26% knowledge.
- Selective retrieval improves both compute efficiency and final success.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

- Indiscriminate knowledge injection is wasteful and often suboptimal for planning agents.
- KnowSelf learns situational self-awareness via lightweight heuristic token labeling on self-explored trajectories.
- KnowSelf (Llama-8B) outperforms strong baselines: 84.33% vs GPT-4o ReAct 76.12% and GPT-4o ExpeL 79.85%.
- Only 15–26% knowledge usage is needed, compared with 100% for KnowAgent/WKM/ExpeL variants.
- The paradigm transfers across scales: Gemma-2B KnowSelf reaches 79.85%, matching GPT-4o ExpeL.

Future directions: richer uncertainty estimation, multi-hop knowledge routing, and transfer to real-world embodied agents.